

Technical note

Experimental investigation of effects of clay content on wave-induced failure of loosely-deposited submarine slopes

Yandong Bi^{a,b}, Shu Zhou^{a,c}, Xiaolin Tan^a, Zhen Guo^{a,d}, Chongqiang Zhu^{a,d}, Jin Sun^{a,*}, Yu Huang^{a,d,*}^a Department of Geotechnical Engineering, College of Civil Engineering, Tongji University, Shanghai 200092, China^b Shandong Provincial Key Laboratory of Marine Environment and Geological Engineering, Ocean University of China, Qingdao 266100, China^c Key laboratory of Mountain Hazards and Earth Surface Process, Institute of Mountain Hazards and Environment, Chinese Academy of Sciences, Chengdu, 610041, China^d State Key Laboratory of Disaster Reduction in Civil Engineering, Tongji University, Shanghai 200092, China

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ABSTRACT

Wave-induced submarine slope failure and landslides threaten coastal communities and offshore infrastructure. In this study, large wave flume tests were conducted to study the effect of clay content (CC) on submarine slope instability under wave loadings. Excess pore water pressure (EPWP) and soil pressure (SP) distributions within a model submarine slope were measured, and the whole process of the model submarine slope losing stability was captured. The experimental results show that the shoulder part of the submarine clayey slope is the first place for liquefaction due to the wave shoaling effect. The peak value of EPWP within a given submarine clayey slope gradually increases vertically, however, the maximum peak EPWP occurs in the middle part of the slope horizontally because of wave scouring and subsequent sliding under reciprocating wave oscillations and gravity. Especially, the results indicated that the CC has a substantial impact on the response of EPWP. When the clayey particles materials are added to the silt, it becomes increasingly susceptible to liquefaction. However, the peak value of EPWP exhibits a nonlinear relationship with CC. There is a critical value, when the CC is greater than this value, the peak EPWP will be partially repressed, which means liquefaction resistance. A possible mechanism about this different susceptibility to liquefaction of the loosely-deposited silty-clay mixtures induced by wave loadings have been proposed: The filling and adhesion effects of fine clay particles influence the localized migration themselves, further influence the EPWP response. This study provides practical guidance for offshore infrastructure design in clay-rich seabeds.

1. Introduction

The liquefaction of loosely deposited saturated soils and seabed instability in submarine environments and the resulting submarine landslides represent not only critical geological processes in seafloor sediment transport but also one of the most significant threats posed by marine geological disasters (Locat and Lee, 2002; Masson et al., 2006; Talling et al., 2014; Huang et al., 2015; Williams, 2016; Vittorio Maselli et al., 2020; Hui et al., 2021; Zhu et al., 2023). And the wave loading is a primary trigger of seabed liquefaction and instability, particularly in coastal and offshore regions (Miyamoto et al., 2004; Ye, 2017; Dou et al., 2021; Lin et al., 2023). The recent advancements in marine resource exploration and marine engineering construction have heightened

concerns regarding wave-induced seabed liquefaction and instability, which may pose significant threats to offshore infrastructure and coastal communities (Randolph and White, 2012; Wang et al., 2018). For example, sinking of breakwaters, overturning of offshore platforms, exposure or floating of submarine pipelines, and other engineering accidents occur from time to time (Bea et al., 1983; Damgaard et al., 2006; Puzrin et al., 2010; Chen et al., 2013; Fan et al., 2023). And the wave-induced liquefaction and instability of seabed has been one of the research focuses in marine geology (Wang et al., 2020; Chen and Xing, 2023). Therefore, it is imperative to investigate the wave-induced seabed liquefaction problems in order to enhance the safety planning and design of marine structures (Wu et al., 2022; Xu et al., 2022).

Wave-induced seabed liquefaction has been studied quite extensively

* Corresponding authors at: Department of Geotechnical Engineering, College of Civil Engineering, Tongji University, Shanghai 200092, China.

E-mail addresses: 23164@tongji.edu.cn (J. Sun), yhuang@tongji.edu.cn (Y. Huang).

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since the 1970 s (Henkel, 1970; Prior and Suhayda, 1979; Prior et al., 1989; Zen and Yamazaki, 1990). It is well established that liquefaction occurs when the excess pore water pressure (EPWP) reaches the vertical effective stresses (Liu et al., 2013; Wang et al., 2020). Two well-documented mechanisms of wave-induced liquefaction exist: momentary (or oscillatory) liquefaction and residual (or accumulated) liquefaction (Sumer, 2014). The former occurs when a wave trough goes by and causes a sharp upward pressure gradient (Sakai et al., 1992; Alcérreca-Huerta and Oumeraci, 2018), whereas the latter is caused by the accumulation of EPWP (Hu et al., 2020; Liu et al., 2022).

Numerical simulation serves as one of the primary approaches for investigating the wave-induced seabed liquefaction and submarine landslides. In previous studies, researchers have carried out comprehensive simulation investigations by finite element method (Ye and Yu, 2021), finite difference method (Zhang and Puzrin, 2021), finite volume method (Duan et al., 2019; Wan et al., 2023), Smoothed Particle Hydrodynamics (Dai et al., 2023), Material Point Method (Dong et al., 2020), and hybrid methods (Xu and Dong, 2021; Sørli et al., 2025). However, due to the limitations of theoretical analysis and numerical simulations on capturing the certain physical characteristics and the whole failure process of seabed, the physical model tests, particularly wave flume experiments designed to capture wave-seabed interactions, have been extensively conducted to investigate the issues of wave-induced liquefaction in cohesionless seabed (such as sandy or silty seabed), mainly focusing on the effect of wave characteristic and sediment properties and wave-structure-seabed interaction problems, etc. (Sassa and Sekiguchi, 1999; Liu et al., 2013, 2017, 2020; Li et al., 2014; Xu et al., 2016, 2022; Zhang et al., 2019; Niu et al., 2019; Wang et al., 2020). Among these studies, for sandy seabed, Sassa and Sekiguchi (1999) conducted an earlier study on the liquefaction of loosely packed sandy seabed under wave loading using centrifuge tests, presenting a critical threshold towards liquefaction. Kudella et al. (2006) performed wave flume experiments to study the pore pressure response in a seabed beneath a caisson breakwater. Furthermore, much progress has been achieved, and the related research could be found in Sumer (2014), Jeng (2018) and Lin et al. (2023). As for silty seabed, scholars have undertaken comprehensive investigations into the silty sediments of the Yellow River Delta (Prior et al., 1989; Jia et al., 2020). For instance, Jia et al. (2014) conducted wave flume tests using silty sediments, and confirmed that waves facilitate sediment re-suspension through liquefaction. Xu et al. (2016) focused on the seabed stratification problem caused by silty sediment liquefaction under storm wave, and found the differentiation phenomenon of variable size deposits. Liu et al. (2017) further proposed an integrated model about this stratification problem. Furthermore, thanks to the development of in situ monitoring technologies and experimental techniques, Wang et al. (2020) proposed a developmental model for seabed instabilities by combined in situ observation and wave flume tests.

However, the loosely-deposited clayey flat or sloping seabed are quite prevalent in natural marine environments (Dou et al., 2021; Wu et al., 2022). For example, in Norway, the clay-covered area amounts to 5000 km², with 20 % of it being strongly structured clay (L'Heureux et al., 2014). And a wide range of clay is distributed in China's sea areas, especially in the South China Sea (Ren et al., 2020; Zhu et al., 2023). This provides conditions for the occurrence of submarine landslide. Even though these studies mentioned above have provided valuable insights that enhance our understanding of the EPWP response and liquefaction mechanisms of cohesionless seabed, to the authors' knowledge, little are known about the instability of submarine clayey slopes induced by wave loadings, and even fewer studies examined the effects of clay content (CC) on instability of submarine slopes (Wit and Kranenburg, 1997; Feng et al., 1992; Gratchev et al., 2006; Bi et al., 2024a). Among the few studies, Kirca et al. (2014) performed wave flume experiments to investigate the influence of CC on clayey seabed liquefaction under progressive waves with silt/sand-clay mixtures, and found that sand or silt may become susceptible to liquefaction when CC up to 30 %. And Liu

and Jeng (2016) further presented a cylinder compressive tests, and their results showed that, when CC falls below a certain value, which varies depending on the types of clay, combinations of clay and sand become prone to liquefaction. In addition, Dou et al. (2021) performed flume tests to investigate the influence of wave shoaling on sloping clayey seabed. What's more, a review of the literature on earthquake-induced liquefaction reveals a critical threshold regarding the influence of fines content on liquefaction resistance. For example, Polito and Martin (2001) found that the liquefaction resistance is controlled by the soil relative density and is independent of the fines content by cyclic triaxial tests. Gratchev et al. (2006) studied the undrained response of normally consolidated clayey soils to cyclic loading by ring-shear apparatus, and found that this threshold is about 7 % ~ 11 %. Furthermore, the threshold obtained by Perlea et al. (1999) is 15 %, while the values obtained by Chen and Zhang (2016), and Huang et al. (2023) are 30 %. Based on comprehensive literature reviews, it is widely acknowledged that there exists a critical value regarding the influence of fines content on the liquefaction resistance of soil. Nevertheless, the accurate determination of this threshold remains a subject of ongoing debate. For this reason, in this study, several large wave flume tests were performed to better understand the mechanism of wave-induced liquefaction and instability of loosely-deposited submarine clayey slopes. This study fills a gap by focusing on the effect of clay content in submarine slope stability, an area with limited experimental research.

The main objective of this work was to reproduce the wave-induced liquefaction and instability process of a submarine slope with high portions of clay content and explore the inherent mechanism that may provide design recommendations for offshore structures. This manuscript is structured as follows: First, a detailed description of test setup is provided, followed by an explanation of the basic characteristics of silt-clay mixtures, and the test procedures and conditions. Subsequently, the wave shoaling effect along the slope was investigated. Thereafter, the accumulation of EPWP and its corresponding patterns in submarine clayey slope models were analyzed. Finally, the effect of clay content was further revealed. This work provides new insights into clay content-dependent liquefaction, crucial for offshore infrastructure safety.

2. Methods

2.1. Model setup and Instrumentation

As illustrated in Fig. 1, the wave flume measures 0.8 m in width, 1.25 m in depth, and 50 m in length, with a constant water depth of 0.5 m maintained throughout this study. A box measuring 0.65 m × 0.5 m × 2.7 m is positioned centrally within the wave flume. The hand-filled, loosely-deposited clayey slope model was situated within this box. According to Hance (2003), most of the submarine landslides occurred on the slope of less than 10 degrees. Thus, the slope model has an inclination angle of 8.5°, with dimensions of 2.7 m in length and 0.3 m in width. The model height gradually decreases from 0.8 m at the shoulder part to 0.5 m at the toe part.

In this study, a separate wave gauge (WG-I) was employed to monitor the far-field wave height before slope model. Additionally, three additional wave gauges (WG-II-IV) were utilized to record the variations in wave heights along the inclined slope surface of the loosely-deposited submarine slope model. These WGs have a measurement range of 50 cm, with a precision of 0.125 mm. Meanwhile, as shown in Fig. 1(d), the soil pressure (SP) and the excess pore water pressure (EPWP) were measured using twelve soil pressure transducers (SPTs) and twelve pore water pressure transducers (PPTs), which were installed along the centerline of the slope at three sections II-IV. These transducers were securely affixed to a steel fixed rack to avoid movement caused by wave loadings during the experiments. However, it should be noted that due to sensor damage or unignorable movement during the experiment, the data of PPT2-1, PPT2-2 and PPT4-1 are invalid and therefore not listed in this paper. Additionally, all transmission cables positioned above the

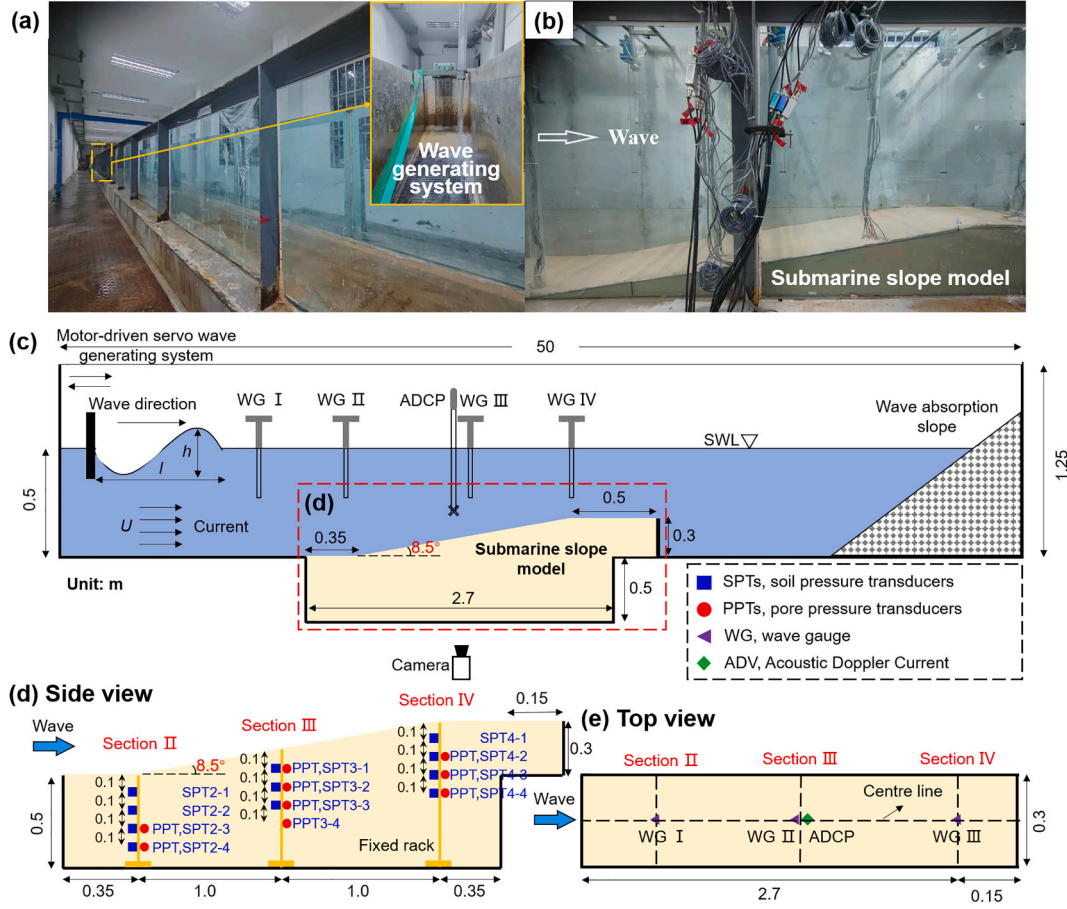


Fig. 1. Photographs of (a) the wave flume and (b) the loosely-deposited submarine clayey slope model; (c) the schematic diagram of the flume; (d) the side view and (e) the top view of the detailed layout of transducers (PPTs, SPTs; WG, wave gauge; and ADV, Acoustic Doppler Current).

slope surface were enclosed within hollow acrylic tubes, which were firmly attached to the brackets. The data were recorded using the iPotest-32S data acquisition instrument with a sampling frequency of 100 Hz. And a high-speed camera was utilized to document the failure process of the loosely-deposited submarine clayey slope model.

2.2. Materials and preparations

The mixtures used in the experiments consists of a composite blend of silt and kaolin clay. Kaolin was selected for its low plasticity, representing fine-grained marine clays. The properties of kaolin clay were characterized by a plastic limit w_p of 11.78 %, a liquid limit w_L of 34.56 %, a plasticity index I_p of 22.78 %. Furthermore, a mixer was used for

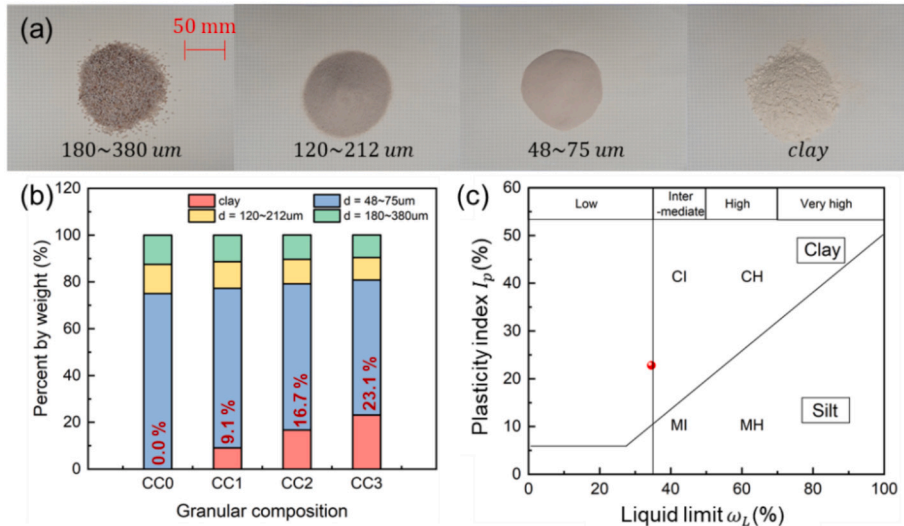


Fig. 2. Soil properties. (a) The photographs of the silt and clay, (b) the weight percentages of the grain size fractions and (c) the soil plasticity diagram of clay.

preparing the silt-clay mixtures. The weight percentages of the grain size fraction of silt-clay mixtures and the clay plasticity diagram are shown in Fig. 2. And the basic properties of these mixtures are summarized in Table 1. The clay content (CC) weight percentages were 0 %, 9.1 %, 16.7 %, and 23.1 %. The density and the water content were measured after consolidation. And the standard falling head method was employed to determine the coefficients of soil permeability k .

2.3. Dimensional analysis

In this study, the Froude number was used the similarity criterion:

$$Fr = \frac{v}{\sqrt{gL_r}} \quad (1)$$

where v is water flow velocity, L_r is a characteristic length. According to the Chinese wave database (CWAVE) as presented by Shi et al. (2019), the maximum significant wave height in the Bohai Sea was about 5 m, while it remained below 10 m in the Yellow Sea. Extreme waves in the East China Sea and South China Sea reached heights of 12–14 m. To replicate typical wave scenarios for the Chinese coast, a geometric similarity ratio of $\lambda_L = 40$ is established based on the capability of wave-current flume. Accordingly, the velocity similarity ratio λ_v is determined to be 6.32 based on Froude's similarity criterion. The time similarity ratio can be expressed as $\lambda_t = \lambda_L / \lambda_v = \sqrt{\lambda_L} = 6.32$. For example, a wave condition characterized by a wave height 8.84 cm and wave period 2.0 s corresponds to the prototype wave of 3.54 m in wave height and 12.64 s in wave period. These prototype wave parameters represent a reasonable wave scenario for the Chinese coast. The wave conditions in the model scale mainly fall into the categories of Stokes II or Stokes III waves (Table 2 and Fig. 3).

2.4. Experimental procedures and Test Conditions

The experiments were conducted according to the following procedure: (1) Transducer installation: The PPTs were thoroughly deaired and subsequently soaked to ensure the absence of air. Subsequently, the PPTs and SPTs were fixed to a steel supporting rack. (2) A carefully controlled silt-clay mixture was mixed to form a slope model. (3) Once the slope model was fully constructed, it was allowed to settle for one day before filling the flume up to a predetermined depth of 0.5 m, followed by a consolidation period lasting around 5–7 days for the submarine loosely-deposited slope model. (4) Subsequently, four series of waves were sequentially generated (see Table 2). (6) Finally, data including pore water pressure, soil pressure, and wave height were recorded using iPotest-32S data acquisition instrument. Table 3 lists all the test conditions of this study.

3. Results

3.1. Wave shoaling

This section analyzes the variations of wave heights along the submarine slope surface to clarify the characteristics of the wave pressure field in this study. Fig. 4 shows the temporal evolutions of the non-dimensional wave height H/H_0 at three sections along slope surface

Table 1
Basic properties of silt-clay mixtures used in the experiment.

Property	CC0	CC1	CC2	CC3
Clay content (CC%)	0	9.1	16.7	23.1
Soil density ρ (g/cm ³)	1.79	2.84	2.75	2.69
Initial void ratio e_0	0.89	0.79	0.63	0.84
Water content (w%)	26.18	24.14	30.50	30.02
Soil permeability k (m/s)	5.04×10^{-4}	3.15×10^{-5}	1.0×10^{-5}	6.49×10^{-6}

Table 2

Details of wave conditions in the flume experiment.

Wave series	Wave height H (cm)	Wave period T (s)	H/gT^2	h/gT^2
1	3.69	2.4	0.00088	0.00885
2	8.84	2.0	0.00255	0.01274
3	14.09	1.6	0.00597	0.01991
4	19.82	1.2	0.01416	0.03539

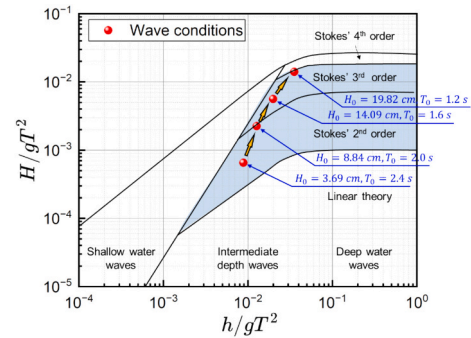


Fig. 3. Wave conditions in the model scale.

Table 3

Details of test plans in the experiment.

No.	ID	Soil type	Clay content (CC %)	Water depth h (cm)	Slope angle ($^\circ$)	Liquefaction?
1	CC0-A8	Silt alone	0	50	8.5	No
2	CC1-A8	Silt-clay mixtures	9.1	50	8.5	Yes
3	CC2-A8	Silt-clay mixtures	16.7	50	8.5	Yes
4	CC3-A8	Silt-clay mixtures	23.1	50	8.5	Yes

of CC2-A8, where H_0 is the wave height measured by WG-I. It shows that, as waves move up along the slope model, their crests become sharper and there is an increase in nonlinearity of wave troughs because of shoaling effects. This phenomenon has been previously discussed by Dou et al. (2021) and Liu et al. (2022).

Fig. 5 further display the variations of non-dimensional wave height H/H_0 along the slope surface at different sections of CC2-A8. Note that the horizontal coordinate x/L_0 represents the relative horizontal distance from section II to section IV (see Fig. 1). As shown in Fig. 5, presumably due to the energy dissipation during the propagation process, the wave height arriving at the toe of the slope is reduced to a certain extent in comparison with the far-field wave height at section I. Besides, for the 3.36 cm, 8.84 cm, and 14.09 cm wave heights, the wave height gradually rises as the wave propagates upwards over the slope. In other words, a remarkable wave shoaling effect occurs along the slope. The wave height near the shoulder is more than twice of that at slope toe. This suggests that the shoulder area is highly susceptible to wave actions and may be considered as the most vulnerable part. The wave height of 19.82 cm did not undergo significant wave shoaling effects due to two factors: (i) the occurrence of intense wave breaking, and (ii) the failure of the slope resulted in a flatter bed as a result of soil liquefaction and subsequent slope failure, which will be comprehensively analyzed in the next section.

3.2. Instability process of submarine slope

Fig. 6 and Fig. 7 shows the snapshots of the failure process of loosely-

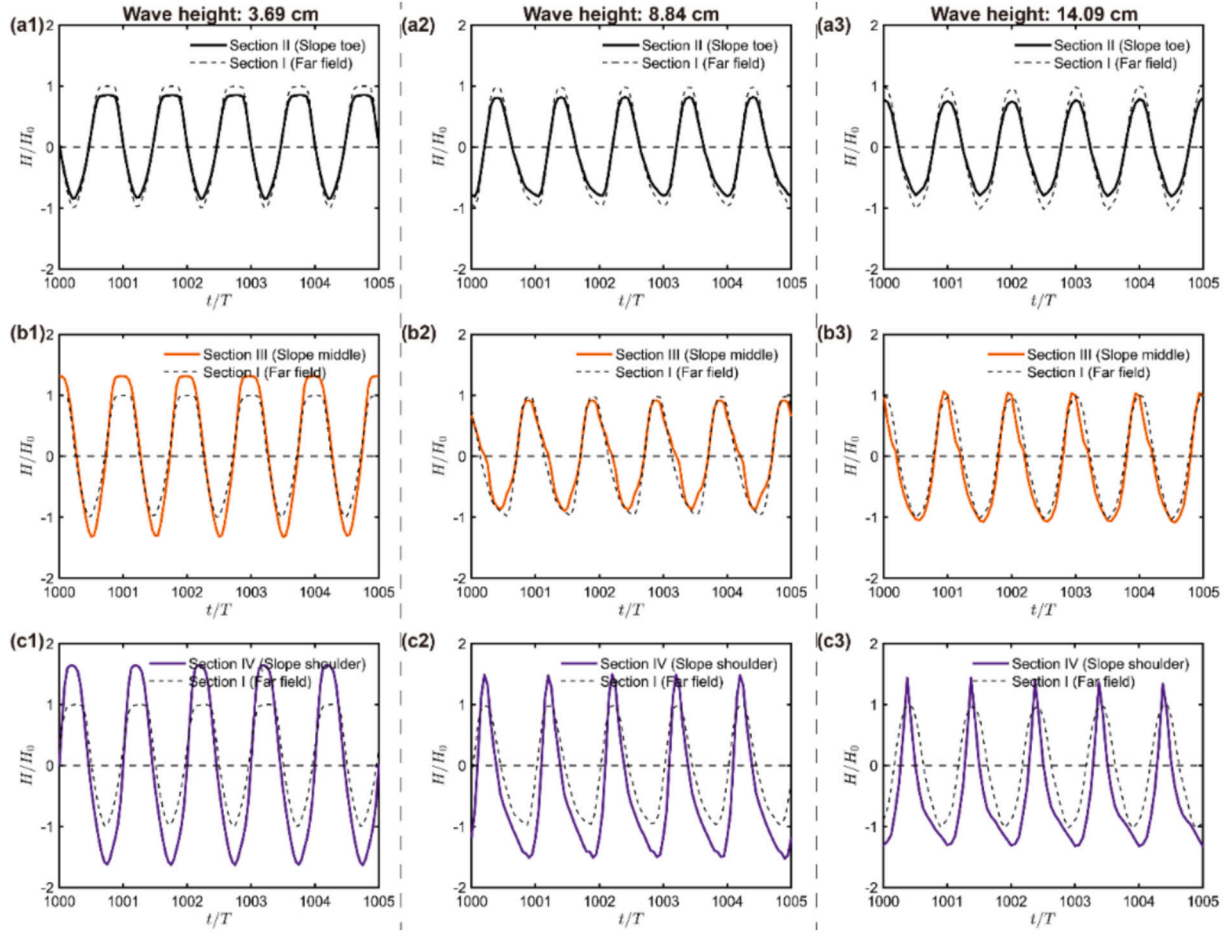


Fig. 4. Time variations of the non-dimensional wave height H/H_0 at three sections of CC2-A8: (a) section II: slope toe; (b) section III: slope middle; (c) section IV: slope shoulder.

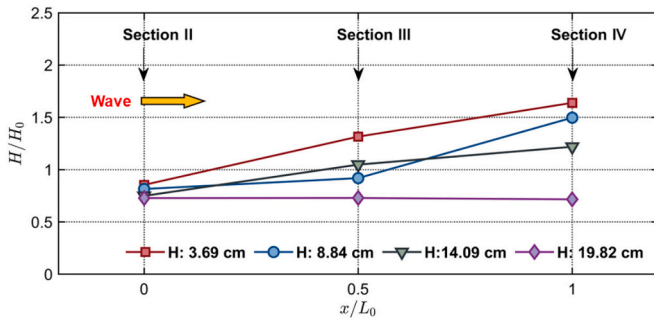


Fig. 5. The variations of wave height along the slope surface from section II to section IV of CC2-A8.

deposited submarine slope model composed of different silt-clay mixtures. It is seen that, for the cases of pure silt, i.e. CC0-A8, the slope surface was reshaped and continuous ripples were generated under progressive wave scouring without obvious failure events (Fig. 6 (a2)). As for the slope composed of silt-clay mixtures, the more detailed failure process, including the settlement curves of seafloor surface, is illustrated in Fig. 7, using CC2-A8 as an example. What's more, the displacement of the liquefied soil and the liquefaction interface of CC1-A8 from PIV analysis are illustrated in Fig. 8. They show no obvious ripples but the slope model failure, the failure process can be outlined as follows:

- (I) Initial shallow surface soil liquefaction of the slope shoulder and subsequent intense scouring. In contrast to the silty slopes, which are prone to scouring (such as CC0-A8), the clayey slopes didn't exhibit significant scouring phenomenon until surface liquefaction taking place at the shoulder part. After initial liquefaction, the silt-clay mixtures endured wave-induced reciprocating oscillation. Meanwhile, the fine clay particles were taken away, leaving behind the coarser particles.
- (II) Sliding of the liquefied soil along the slope. Concurrently, the residual coarser sand particles descended towards the toe of the slope, as illustrated by the dark-colored sliding mass in Fig. 6(b2, c2 and d2), and the steep increase of the settlement curves of the slope toe in Fig. 7(b).
- (III) Deeper liquefaction and further sliding. Simultaneously, the wave pressure propagated deeper into the soil, resulting in an increasing liquefaction depth and the formation of an arc-like liquefaction interface (Fig. 6(b2, c2, d2) and Fig. 8). The liquefied soil was moved further along the liquefaction interface, and slid down towards the toe of the slope (Fig. 7). With the development of liquefaction, more and more mixtures slid down the slope and deposit at the slope toe, finally forming a flat seabed (Fig. 6(b3, c3, d3) and Fig. 7(a).

3.3. Pore water pressure response

As previously stated, when a wave travels up a slope, it becomes steeper due to the shoaling effect. This can result in an increased wave pressure on the surface near shoulder part, which differs from that

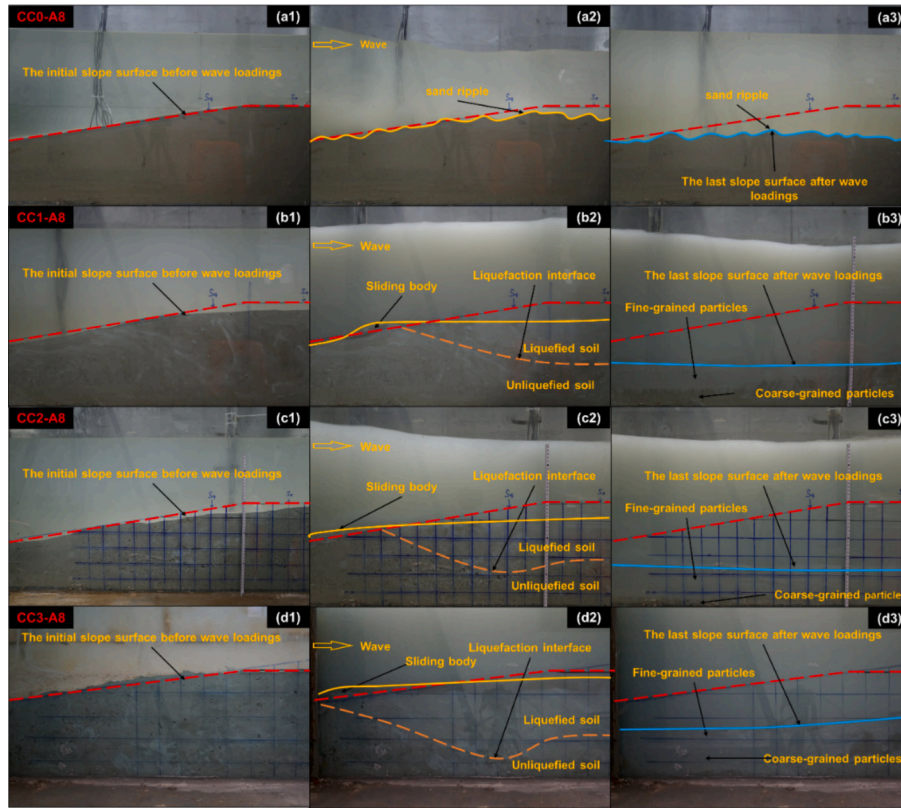


Fig. 6. Typical photographs of instability process of submarine slope of (a) CC0-A8; (b) CC1-A8; (c) CC2-A8; (d) CC3-A8.

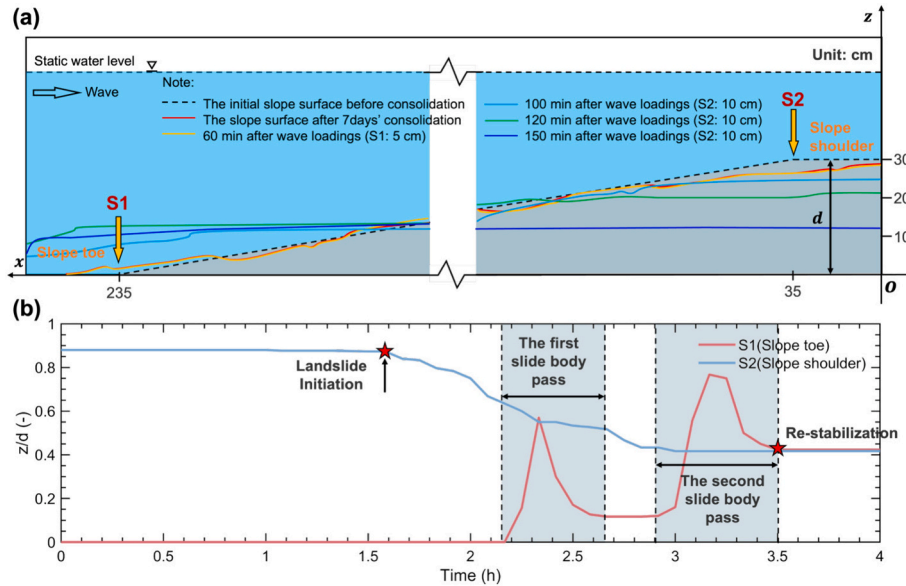


Fig. 7. Slope surface in different times of CC2-A8. (a) settlement curves of seafloor surface, and (b) relative heights of S1 and S2 on seabed surface.

experienced on a flat seabed. As a consequence, there may be variations in the transient pore water response of the submarine slope and this can further affect its stability. In this section, we will examine the EPWP response, leaving the corresponding soil pressure responses to the next section.

Fig. 9 shows the EPWP evolution at different section and buried depth within CC2-A8 slope model. All of the lines exhibit a consistent evolutionary pattern: initially, the EPWP gradually builds up until it approaches a critical threshold, leading to soil initial liquefaction;

subsequently, the liquefaction progresses downwards, following a compaction process. This typical wave-induced liquefaction sequence has been reported by Sumer et al. (2006) and further reviewed by Jeng (2013) and Sumer (2014).

Specifically speaking, under the loading of a small wave height of 3.69 cm (Stage: S1), as shown in Fig. 9, there is almost no build-up of EPWP at the toe and middle part. This suggests that the wave pressure can hardly push the toe and the middle part of the slope under this relatively small progressive wave. However, due to obvious wave

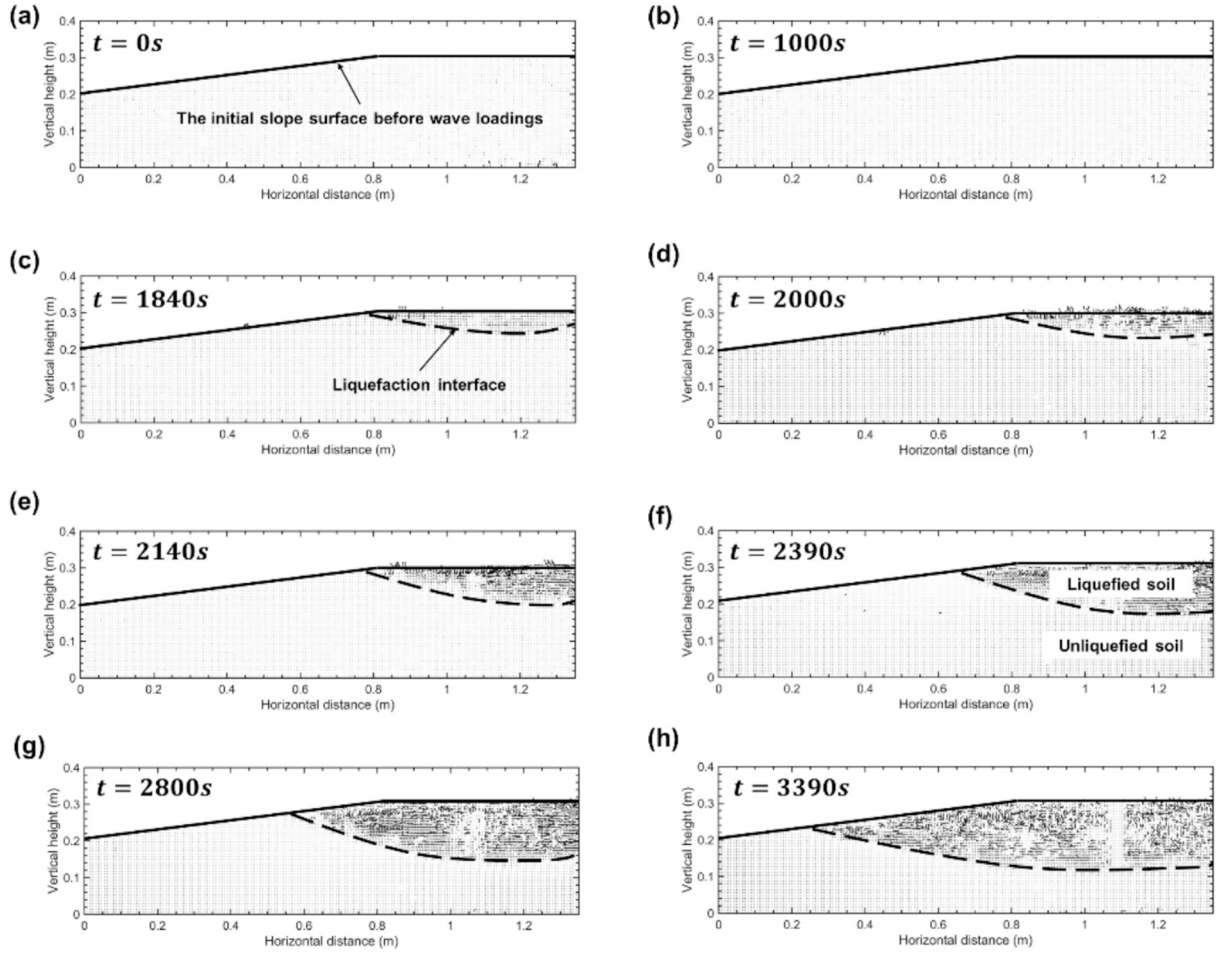


Fig. 8. Displacement vector and liquefaction interface of submarine clayey slope of CC1-A8 from different PIV analysis time.

shoaling (see Fig. 5) at the slope shoulder, there is a smaller accumulation of EPWP observed at this location (as depicted in Fig. 9(c)).

Upon the application of larger progressive waves (Stage: S2), EPWP gradually accumulated after several wave cycles. Subsequently, the EPWP significantly increased, reaching its peak value, which indicates that it had approached the initial liquefaction state at this location (Sumer et al., 2006; Zhao et al., 2017; Liu et al., 2022; Wang et al., 2020). After liquefaction in the shallow surface soils, the EPWP would further transmit and accumulate from shallow to deeper soil, led to progressive liquefaction (Sumer, 2014). This can be demonstrated by the time taken for the EPWP to reach its peak value at different depths. After continuing to impose larger progressive wave (Stage: S3&S4), the EPWP dissipated continuously instead of further built-up, named as compaction process. Furthermore, it is evident that the peak EPWP increases gradually with depth.

Fig. 10 presents the wave-induced peak EPWP at the same depth under the surface of the CC2-A8 slope model. As depicted in Fig. 10, the peak EPWP does not increase gradually in the direction of the slope, but rather reaches its maximum value at section III, which is located in the middle of the slope. This is because while liquefaction is occurring on the slope surface, scouring and sliding of the liquefied soils will occur subsequently. It's widely known that wave loads can lead to EPWP buildup, but wave erosion and scouring promote the dissipation of EPWP (Dou et al., 2021; Liu et al., 2022). As a result of the continuous wave action, the cyclic waves would scour the soil on the slope shoulder. Simultaneously, the shallow surface layer of the already liquefied soil would gradually slide down due to oscillation of waves. This sliding process leads to a decrease in overlying soil thickness and therefore

reduces overlying effective stress. Therefore, the actual EPWP generated at the shoulder of the slope will be lower than the value without considering the scouring and sliding of liquefied soil layers at the shoulder. On the contrary, wave-induced EPWP at the slope toe would be larger than the ideal conditions. This is due to the deposition of landslide bodies at slope toe (Fig. 7) resulting in an increased overlying effective stress and a longer dissipation path.

In order to further analyze the effects of these two components on the stability of loosely-deposited submarine clayey slopes, the moving average method with a window size of 2000 was used for EPWP signal filtering to obtain the OPWP and RPWP components. Fig. 11 shows the time series of wave-induced OPWP and RPWP at slope middle (section III) of CC2-A8. For oscillatory pore water pressures (OPWP), under wave loadings of Stage S1, the amplitude of OPWP varies greatly along depth direction, and presents a decline pattern. For example, when the depth is 10 cm (i.e. PPT3-1), the normalized OPWP amplitude $|p/p_0|$ is 0.23. And the $|p/p_0|$ at depths of 20 cm, 30 cm, and 40 cm are 0.21, 0.07 and 0.05, respectively (Table 4). There, the wave pressure on the surface of the slope p_0 is obtained from Eqs. 2 and 3 based on small amplitude wave theory:

$$p_0 = \frac{\gamma_w H}{2 \cosh(2\pi h/L)} \cos(kx + wt) \quad (2)$$

$$\frac{H}{H_0} = \left(\frac{k}{k_0} \frac{1 + \frac{k_0 h_0}{\sinh(k_0 h_0)}}{1 + \frac{kh}{\sinh(kh)}} \right)^{1/2} \quad (3)$$

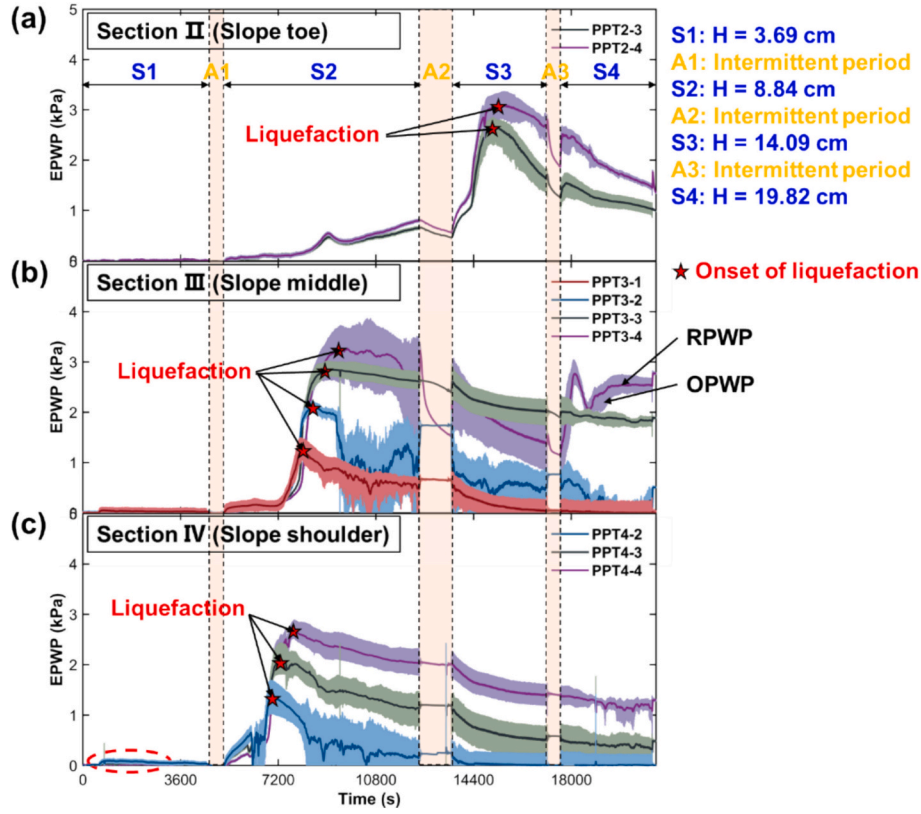


Fig. 9. Time series of wave-induced excess pore water pressures (EPWP) at vertical locations of different sensors on the submarine slope model of CC2-A8. (a) section II is at the toe of the slope; (b) section III is at the middle of the slope; (c) section IV at the shoulder of the slope. Note: RPWP represents residual pore water pressure components, and OPWP represents the oscillatory pore water pressure components. S1-S4 represent the periods of generating wave, and A1-A3 represent the intervals between two adjacent generating-wave periods.

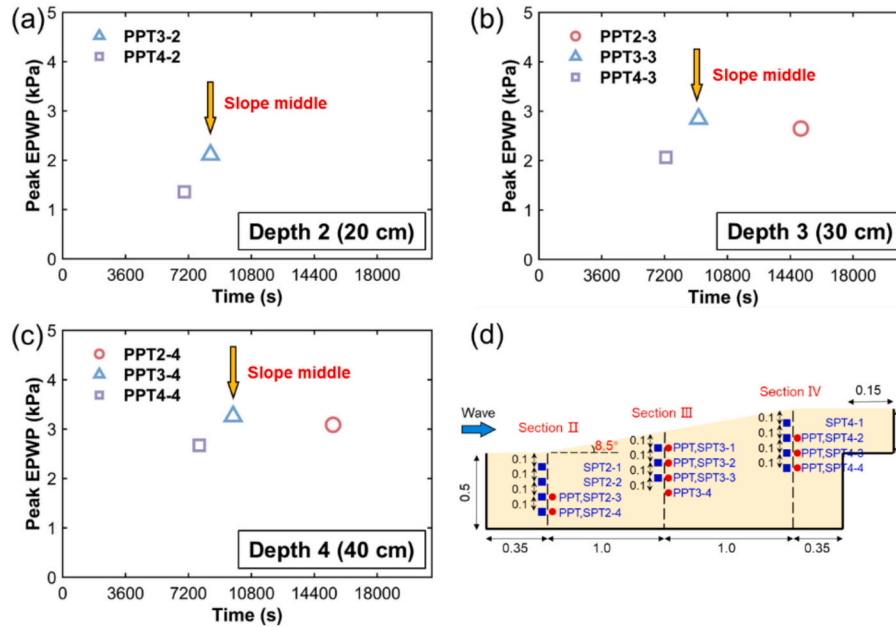


Fig. 10. Wave-induced peak excess pore water pressures at the same depth under slope surface on the submarine slope model of CC2-A8. (a) PPTs at 20 cm under slope surface, PPT2-2 is broken; (b) PPTs at 30 cm under slope surface; (c) PPTs at 40 cm under slope surface; (d) the detailed layout of transducers.

where γ_w is water unit weight, h_0 is mean still water depth, h is water depth at slope surface, H_0 is wave height in the far-field of the slope (see Table 2), H is wave height at slope surface. w is wave frequency, k_0 and k are the wave numbers before and after wave climbing, respectively.

However, under wave loadings of Stage S2, the OPWP amplitude exceeds p_0 . This amplification effect of PWP may be related to the liquefaction and fluidization oscillation of post-liquefaction. Besides, it is found that the envelope lines of OPWP shows different pattern,

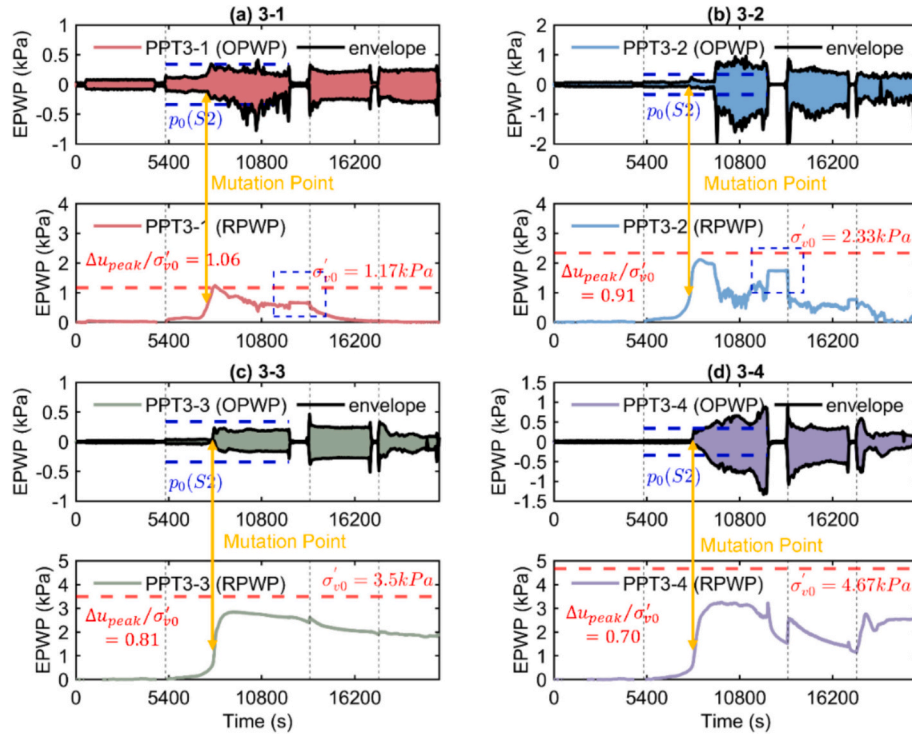


Fig. 11. Time series of wave-induced oscillatory pore water pressures (OPWP) and residual pore water pressure (RPWP) at middle part of submarine slope model of CC2-A8. (a) PPT3-1 (10 cm); (b) PPT3-2 (20 cm); (c) PPT3-3 (30 cm); (d) PPT3-4 (40 cm). Note: $\sigma'_{v0} = \gamma'z(1 + 2k_0)/3$ is the initial mean normal effective stress, where $\gamma' = (\rho - \rho_w)g$ is the submerged specific weight of the soil, $k_0 = 0.5$.

Table 4
EPWP response of PPT3 of CC2-A8.

PPT	$ p/p_0 $ at Stage 1	Liquefaction?	Shape of envelope lines	σ'_{v0}	$\tau_{u,peak}$
3-1	0.23	Yes	Comb-tooth type	1.17	1.06
3-2	0.21	Yes	Comb-tooth type	2.33	0.91
3-3	0.07	Yes	Stepped type	3.50	0.81
3-4	0.05	Yes	Violin type	4.67	0.70

including comb-tooth type (PPT3-1&3-2), stepped type (PPT3-3) and violin type (PPT3-4). These patterns may be related to the depth and the degree of soil consolidation. The shallower soil underwent liquefaction under continuous cycle wave loadings, and then became a two-layer fluid system which had a reciprocating oscillation behavior governed by wave loadings, corresponding to a comb-tooth type. As for the deeper soil, the OPWP pattern would behave as a stepped type. The occurrence of violin type may be due to soil under-consolidation. And it is worth noting that the envelope lines of OPWP could be taken as a qualitative early warning indicator to judge the occurrence of liquefaction in submarine clay slope. As shown in Fig. 11, the mutation point of the envelope line of the OPWP curve corresponds to the vicinity of the maximum value of the slope of the RPWP curve. This indicates that liquefaction and instability will occur on the submarine slope in a relatively short period of time. In addition, as shown in the dashed blue rectangular box in Fig. 11(a, b), it also can be found that the RPWP increases significantly after S2 is stopped. This is probably because when wave loadings stopped, the upper liquefied soil would deposit again. At this time, the body weight of the soil is completely borne by the pore water pressure, so the RPWP increases abruptly.

3.4. Soil pressure response

Fig. 12 shows the time series of soil pressures (SP) at middle part of loosely-deposited submarine clayey slope model of CC2-A8. It was found

that vertical effective soil pressure of all four time-traces show a similar small degree of enlargement under the action of a relatively small progressive wave (Stage: S1). It indicates the smaller progressive wave has a good pre-consolidation effect for these loosely-deposited clayey slope (Zhang et al., 2019). After that, the time traces of EPWP curve will gradually increase to a maximum value and then remain stable for a period of time. Corresponding to this, the SP would rapidly reduce from the maximum value to the minimum value in a short period of time. However, the SP of SPT3-2, SPT3-3 and SPT3-4 show a certain degree of recovery after the descending segment. This indicates that some of the liquefied soils have been reconsolidated. In addition, it's also can be found that the time when the mutation point of the SP-time history curve appears does not correspond to the peak point of the EPWP-time history curve, but the vicinity of the maximum value of the slope of the RPWP curve. In other words, the mutation point of the SP-time history curve corresponds to the mutation point of the envelope line of the OPWP curve (Fig. 11). This further substantiates the rationale for selecting this point as the early warning indicator.

4. Discussion

4.1. Effects of clay content

The effects of CC on wave-induced EPWP in the loosely-deposited submarine slope model are clearly displayed in Fig. 13, where the peak values of wave-induced RPWP are extracted separately to plot a scatter graphic against CC. To facilitate comparison, we redefine the moment when applying the wave with 8.84 cm wave height as the zero moment. This is because all PPTs did not show significant built-up of EPWP before this moment. As can be seen, all the silt-clay mixtures exhibited a clear EPWP built-up phenomenon except for the silt-only case, i.e. CC0-A8. This indicates that the addition of a certain amount of clay increases the mixture's susceptibility to liquefaction, as it decreases the distance to the critical EPWP ratio after the same number of

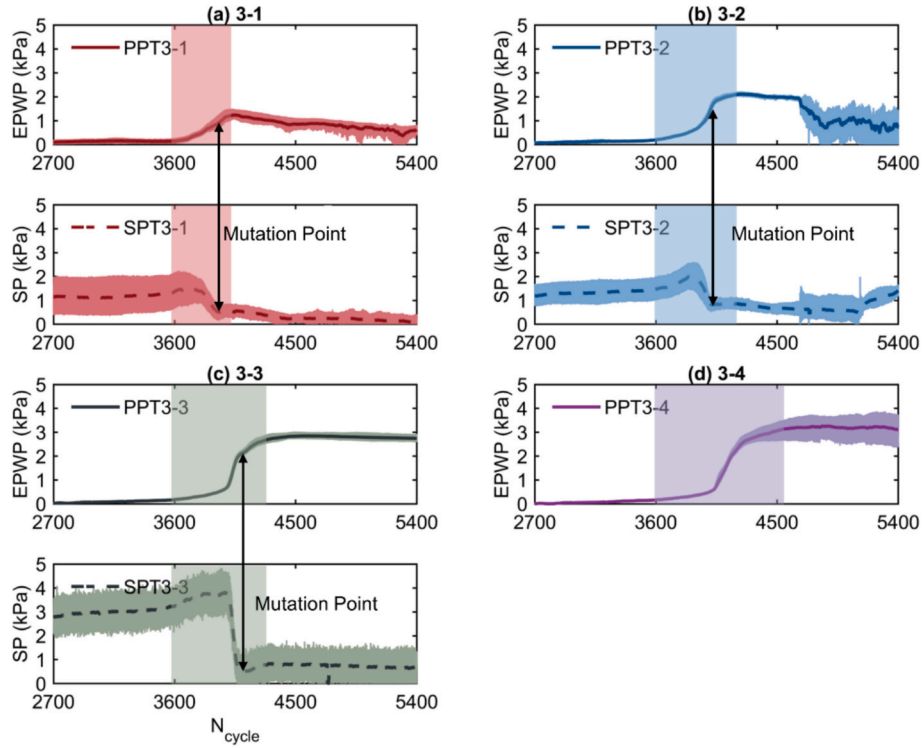


Fig. 12. Time series of excess pore water pressure and the corresponding soil pressures at middle part of submarine slope model of CC2-A8. (a) SPT3-1 (10 cm); (b) SPT3-2 (20 cm); (c) SPT3-3 (30 cm); (d) SPT3-4 (40 cm).

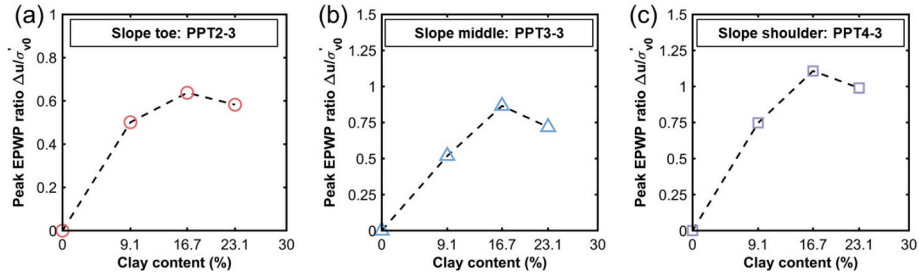


Fig. 13. Wave-induced EPWP and the corresponding peak EPWP ratio at different locations on the submarine slope model for different clay content. (a) Slope toe, (b) slope middle, and (c) slope shoulder.

loading cycles. However, it is also obvious that the peak value of EPWP increases with rising CC for CC smaller than about 16.7 %; however, for CC greater than 16.7 %, the submarine slope generates a relatively small peak value. It indicates that the further increasing of clay content would reduce the susceptibility of silt to liquefaction, that is to say, increase the liquefaction resistance. In other words, the liquefaction susceptibility of silt-clay mixture exhibits an initial increase followed by a subsequent decrease as the clay content increases, indicating the presence of a critical value for clay content.

To further substantiate this conclusion, undrained cyclic triaxial tests were additionally conducted using the same silt-clay mixtures as those used in the flume tests. The dynamic triaxial experiments were conducted by GDS dynamic triaxial apparatus of Tongji University. The specific operation procedures and cautions are not elaborated in this article. The detailed test conditions are summarized as follows: Take CC2 of Table 1 as an example, the sample size was 38 mm in diameter and 80 mm in height. The initial effective confining pressure of the specimen is 275.15 kPa, which represents a representative volume element under an overburden thickness of 16 m:

$$\sigma'_{c0} = \gamma' z = (\rho - \rho_w)gz \quad (4)$$

The cyclic stress ratio ($CSR = \sigma_d / 2\sigma'_{c0}$, where σ_d is the deviatoric stress) adopted there is 0.15.

The waveforms adopted in this experiment was sine waves. And the wave period is 12.64 s, corresponding to a frequency of 0.079 Hz. This condition is consistent with the prototype condition in the flume experiment where $H_0 = 8.84$ cm and $T_0 = 2.0$ s. These tests aimed to investigate the influence of clay content on EPWP dynamics response and liquefaction susceptibility, with the results presented in Fig. 14. As can be seen, the number of cycles required for liquefaction, when the EPWP reaching the mixture's effective stress (Fig. 14(a)), initially decreases and then increases as the clay content increases. This is in line with the trend of liquefaction susceptibility observed in the flume test, indicating an initial increase in liquefaction susceptibility with increasing clay content, followed by a subsequent decrease.

Based on the analysis of instability process of submarine slope and the EPWP response mentioned above, the different susceptibility to liquefaction or failure of this loose silty-clay mixtures induced by wave loadings can be explained as follows (see Fig. 15). When a wave trough passes the loosely-deposited submarine clayey slope model, an upward pressure gradient force would generate the upward seepage in the shallow soil layer. In loose silty-clay mixtures with low CC, the primary

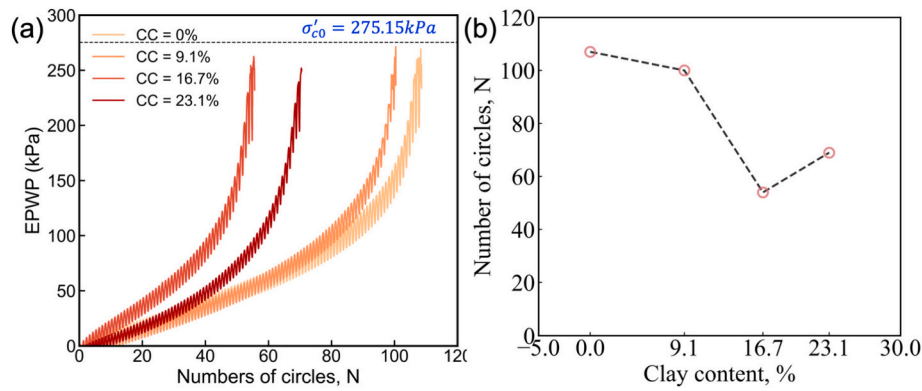


Fig. 14. Variation of number of circles with (a) EPWP and (b) clay content.

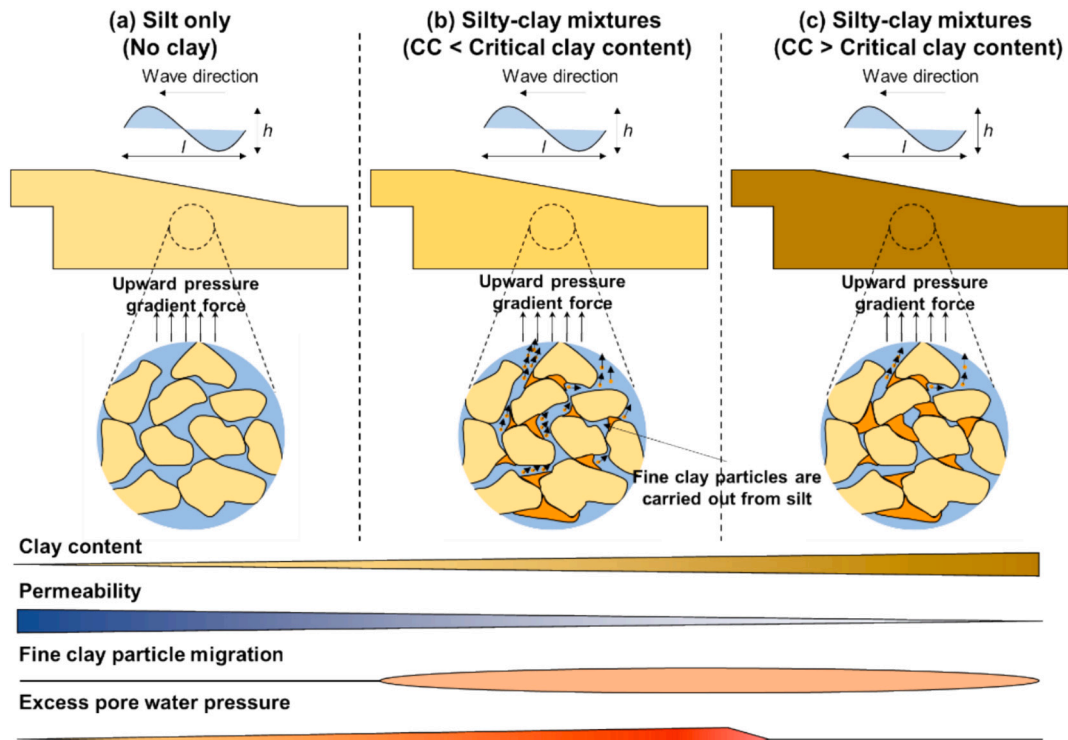


Fig. 15. Schematic representation of the non-linear effects of clay content on EPWP and liquefaction: (a) submarine silty slope, (b) submarine clayey slope with lower CC, and (c) submarine clayey slope with larger CC.

role of fine clay particles is to effectively occupy the void within silt particles, leading to lower permeability as detailed in Table 1. Under cyclic wave loadings, fine clay particles may gradually detach from the silt particle skeleton, as evidenced by the turbid water in Fig. 6. This process results in rearrangement of coarser particles under cyclic shear strains and leads to accumulation of EPWP at the low permeability. Conversely, for loose silty-clay mixtures with high CC, fine clay particles may primarily serve an adhesive function. As a result, these fine clay particles are difficult to dislodge from silt particles and partially or completely inhibit the accumulation of EPWP, ultimately resulting in a lower peak EPWP and more numbers of cycles for liquefaction (see Fig. 13 and Fig. 14). Therefore, the soil structural rearrangement is first facilitated and then inhibited by the CC increase. It should be noted that the decrease in permeability provides an undrained condition and promotes liquefaction. This, however, would not alter the liquefaction behavior at high CC, as evidenced by the fact that pure clay does not liquefy under the undrained condition (Zhang et al., 2007).

4.2. Limitations

Unfortunately, the mobility of localized fine clay particles under wave loadings was not be quantitatively captured in this wave flume experiments. Therefore, further exploration in the future is warranted, such as the utilization of highly accurate PIV, and even the Micro PIV technology (Zhou et al., 2024). What's more, multi-scale numerical simulation methods, such as CFD-DEM, LBM-DEM, etc., provide the possibility of capturing this process in a refined manner (Mu et al., 2023; Bi et al., 2024b; Ma et al., 2024).

It is worth noting that the determination of the critical clay content value holds significant importance for both theoretical research and practical engineering in addressing soil liquefaction issues. However, it should be noted that this experiment only focused on low plasticity kaolin and prepared a limited number of scaled submarine slope models with varying clay contents. Additionally, the soil exhibits a relatively loose initial state (see Table 1). Hence, the universality of the critical

clay content obtained from these test results should be confirmed through a comprehensive array of tests covering various clay type, clay contents and different relative density.

5. Conclusions

This paper presents the results of several large wave-flume tests to better understand the mechanisms of the wave-induced liquefaction and instability of loosely-deposited clayey submarine slopes. The clay content (CC) was varied among the tests and their effects on the EPWP response and the instability of model slopes under wave actions were compared and discussed. The main observations and conclusions are presented as follows.

1. Obvious wave shoaling phenomenon was observed. The wave shoaling makes the shoulder part of loosely-deposited submarine clayey slope the prime place for liquefaction and instabilities.
2. Unlike submarine sandy or silty slopes, the instability process of submarine clayey slope can be divided into three stages: (i) Under wave loadings, the EPWP gradually increases, and the liquefaction firstly occurs near the shoulder part of a submarine clayey slope. The liquefied soil underwent reciprocating oscillation governed by the wave loadings. (ii) The liquefied soil would slide down towards to the toe of the slope. (iii) With the development of liquefaction, more and more mixtures slid down the slope and deposited down at the slope toe, finally forming a flat seabed.
3. For a given submarine clayey slope, the peak value of EPWP at a given section gradually increase along the depth direction. As for the horizontal direction, a maximum value of peak EPWP occurs in the middle part because of wave scouring and the subsequent sliding under wave reciprocating oscillations and the gravity near shoulder.
4. Clay content (CC) significantly affects the EPWP response and the stability of submarine slope. The liquefaction susceptibility of silt-clay mixture exhibits an initial increase followed by a subsequent decrease as CC increases, indicating the presence of a critical value for clay content.
5. A potential mechanism has been proposed to explain this non-monotonic variation of the liquefaction susceptibility of loosely-deposited silty-clay mixtures. It is centered on the idea that the structural rearrangement was first facilitated and then inhibited by the clay addition. The filling and adhesive effects of fine clay particles might result in this different susceptibility to liquefaction. Of course, future studies should test different clay minerals, relative density, slope geometries, and extreme wave conditions.

CRediT authorship contribution statement

Yandong Bi: Writing – original draft, Methodology, Conceptualization. **Shu Zhou:** Visualization, Software. **Xiaolin Tan:** Methodology, Investigation. **Zhen Guo:** Writing – review & editing, Supervision. **Chongqiang Zhu:** Writing – review & editing, Validation, Supervision. **Jin Sun:** Supervision. **Yu Huang:** Writing – review & editing, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

All data, models, and code generated or used during the study appear in the submitted article.

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